

Fuzzy Set

1. What is an Embedded System? Write its Applications.

An **Embedded System** is a computer system designed to perform a **specific task** within a larger system. It consists of **both hardware and software** and is built for a particular application.

Applications of Embedded System

1. Washing Machine
2. Microwave Oven
3. ATM Machine
4. Digital Camera
5. Smart TV
6. Mobile Phone
7. Airbag System in Cars
8. Medical Devices (ECG, Pacemaker)

Example::A **washing machine** is an embedded system because it is designed only for washing clothes automatically.

2. Explain Fuzzy Set and Embedded System.

An **Embedded System** is a computer system designed to perform a **specific task** within a larger system.

A **Fuzzy Set** is a set in which an element can belong **partially** to the set. Each element has a **membership value** between **0 and 1**.

Unlike a classical set, membership is **not only 0 or 1**.

Example::Suppose the fuzzy set "**Tall People**".

Person Height Membership Value

- | | | |
|---|--------|-----|
| A | 150 cm | 0.2 |
| B | 170 cm | 0.6 |
| C | 185 cm | 1.0 |

Here, person **B** is partially tall because the membership value is **0.6**.

3. Explain Fuzzy Logic and Embedded System.

Fuzzy Logic is a reasoning method that works with **degrees of truth** instead of only **True** or **False**. It uses values between **0** and **1**.

Example: Temperature control in an **air conditioner**.

Instead of only **Hot** or **Cold**, fuzzy logic uses values like:

- Very Cold
- Cold
- Warm
- Hot
- Very Hot

Fuzzy Logic in Embedded System

Fuzzy Logic is used in embedded systems to make **smart and automatic decisions**.

Applications

- Washing Machine
- Air Conditioner
- Automatic Car Control
- Smart Camera
- Medical Equipment

4. Difference between Fuzzy Set and Classical Set.

Fuzzy Set	Classical Set
Membership value ranges from 0 to 1	Membership value is only 0 or 1
Partial membership is allowed	Partial membership is not allowed
Used in uncertain problems	Used in exact problems
Based on fuzzy logic	Based on Boolean logic
More flexible	Less flexible

5. Operations on Fuzzy Set and Classical Set.

Classical Set Operations

1. Union ($\cup$)
2. Intersection ($\cap$)
3. Complement (A')

Fuzzy Set Operations

1. Union
2. Intersection
3. Complement

Unlike classical sets, fuzzy operations use **membership values (0–1)**.

6. Representation of Fuzzy Set.

A **Fuzzy Set** is represented by its **membership function**, where every element has a membership value between **0 and 1**.

General Form $A = \{(x, \mu_A(x)) \mid x \in X\}$

Where,

- **A** = Fuzzy Set
- **x** = Element
- **$\mu_A(x)$** = Membership value of x
- **X** = Universal Set

Example $A = \{(10,0.2), (20,0.5), (30,0.8), (40,1.0)\}$

Here,

- 10 belongs to the set with membership **0.2**
- 20 belongs with **0.5**
- 30 belongs with **0.8**
- 40 belongs with **1.0**

7. Membership Function.

A **Membership Function** is a function that assigns a membership value between **0 and 1** to every element of a fuzzy set.

Properties

- Value lies between **0 and 1**.
- **0** means the element does not belong to the set.
- **1** means the element fully belongs to the set.
- Values between **0 and 1** indicate partial membership.

Example :For the fuzzy set **Tall People**:

Height Membership

150 cm 0.2

170 cm 0.6

185 cm 1.0

Properties of Fuzzy Set

A **Fuzzy Set** has several properties based on membership values between **0 and 1**.

Properties

1. Support

- The set of all elements whose membership value is greater than 0.
- **Example:** If $\mu(x)=0.4$, then x belongs to the support.

2. Core

- The set of all elements whose membership value is 1.
- **Example:** If $\mu(x)=1$, then x belongs to the core.

3. Height

- The highest membership value in a fuzzy set.
- **Example:** If the maximum membership value is 1, then the height is 1.

4. Normal Fuzzy Set

- A fuzzy set is normal if at least one element has membership value 1.

5. Convex Fuzzy Set

- A fuzzy set is convex if every point between two elements has a membership value not smaller than the minimum of those two elements.

9. Union of Fuzzy Set

The **Union** of two fuzzy sets contains the maximum membership value of each element.

Formula $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$

Example

x	A	B	A ∪ B
1	0.3	0.6	0.6
2	0.8	0.5	0.8
3	0.4	0.9	0.9

10. Logical Operations in Fuzzy Set

Logical operations are used to perform operations on fuzzy sets using membership values.

Types

AND

Uses **minimum** value.

$$\min(\mu_A, \mu_B)$$

OR

Uses **maximum** value.

$$\max(\mu_A, \mu_B)$$

NOT

Uses complement.

$$1 - \mu_A$$

Example

Suppose,

$$A = 0.7$$

$$B = 0.4$$

$$\text{AND} = 0.4$$

$$\text{OR} = 0.7$$

$$\text{NOT}(A) = 0.3$$

11. Intersection of Fuzzy Set

The **Intersection** of two fuzzy sets contains the minimum membership value of each element.

Formula $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$

Example

x	A	B	A ∩ B
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1	0.3	0.6	0.3
---	-----	-----	-----

2	0.8	0.5	0.5
---	-----	-----	-----

3	0.4	0.9	0.4
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12. Algebraic Operations of Fuzzy Set

Algebraic operations combine fuzzy sets using algebraic formulas instead of minimum and maximum operators.

Types

Algebraic Product $\mu_{A \cap B}(x) = \mu_A(x) \times \mu_B(x)$

Algebraic Sum $\mu_{A \cup B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x)\mu_B(x)$

Example

Suppose,

$$A = 0.6$$

$$B = 0.5$$

Algebraic Product $0.6 \times 0.5 = 0.30$

Algebraic Sum

$$\begin{aligned} &0.6 + 0.5 - (0.6 \times 0.5) \\ &= 1.1 - 0.30 \\ &= 0.80 \end{aligned}$$

Math-1: Union of Fuzzy Set

Question

Let,

$$A = \{(a, 0.2), (b, 0.6), (c, 0.8)\}$$

$$B = \{(a, 0.5), (b, 0.4), (c, 0.7)\}$$

Find the **Union (A $\cup$ B)**.

Solution

Formula: $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$

Element	A	B	Max
a	0.2	0.5	0.5
b	0.6	0.4	0.6
c	0.8	0.7	0.8

Answer $A \cup B = \{(a, 0.5), (b, 0.6), (c, 0.8)\}$

Math-2: Intersection of Fuzzy Set

Find the **Intersection (A $\cap$ B)** using the above fuzzy sets.

Solution

Formula $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$

Element	A	B	Min
a	0.2	0.5	0.2

Element	A	B	Min
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b	0.6	0.4	0.4
---	-----	-----	-----

c	0.8	0.7	0.7
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Answer $A \cap B = \{(a, 0.2), (b, 0.4), (c, 0.7)\}$

Math-3: Complement

Find the **Complement (A')**.

$$A = \{(a, 0.3), (b, 0.5), (c, 0.8)\}$$

Solution

Formula $A' = 1 - \mu_A(x)$

Element	A	1-A
---------	---	-----

a	0.3	0.7
---	-----	-----

b	0.5	0.5
---	-----	-----

c	0.8	0.2
---	-----	-----

Answer $A' = \{(a, 0.7), (b, 0.5), (c, 0.2)\}$

Math-4: Algebraic Product

Find the **Algebraic Product**. $A = \{(a, 0.4), (b, 0.8), (c, 0.6)\}$

$$B = \{(a, 0.5), (b, 0.3), (c, 0.9)\}$$

Solution

Formula $\mu = \mu_A \times \mu_B$

Element	A	B	Product
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a	0.4	0.5	0.20
---	-----	-----	------

Element	A	B	Product
---------	---	---	---------

b	0.8	0.3	0.24
---	-----	-----	------

c	0.6	0.9	0.54
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Answer $A \times B = \{(a, 0.20), (b, 0.24), (c, 0.54)\}$

Math-5: Algebraic Sum

Find the **Algebraic Sum**.

Solution

Formula $\mu = \mu_A + \mu_B - \mu_A \mu_B$

For a

$$\begin{aligned}
 &0.4 + 0.5 - (0.4 \times 0.5) \\
 &= 0.9 - 0.2 \\
 &= 0.7
 \end{aligned}$$

For b

$$\begin{aligned}
 &0.8 + 0.3 - (0.8 \times 0.3) \\
 &= 1.1 - 0.24 \\
 &= 0.86
 \end{aligned}$$

For c

$$\begin{aligned}
 &0.6 + 0.9 - (0.6 \times 0.9) \\
 &= 1.5 - 0.54 \\
 &= 0.96
 \end{aligned}$$

Answer $\{(a, 0.70), (b, 0.86), (c, 0.96)\}$

Math-6: Max-Min Composition

Question

Given $R = \begin{bmatrix} 0.2 & 0.8 \\ 0.6 & 0.5 \end{bmatrix}$

And $S = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.9 \end{bmatrix}$

Find the **Max-Min Composition ($R \circ S$)**.

Solution

Formula $(R \circ S)_{ij} = \max\{\min(R_{ik}, S_{kj})\}$

(1,1) $\max(\min(0.2, 0.7), \min(0.8, 0.3))$
 $= \max(0.2, 0.3) = 0.3$

(1,2) $\max(\min(0.2, 0.4), \min(0.8, 0.9))$
 $= \max(0.2, 0.8) = 0.8$

(2,1) $\max(\min(0.6, 0.7), \min(0.5, 0.3))$
 $= \max(0.6, 0.3) = 0.6$

(2,2) $\max(\min(0.6, 0.4), \min(0.5, 0.9))$
 $= \max(0.4, 0.5) = 0.5$

Answer $R \circ S = \begin{bmatrix} 0.3 & 0.8 \\ 0.6 & 0.5 \end{bmatrix}$